



Do what matters

The Speech Emotion Recognition Agent

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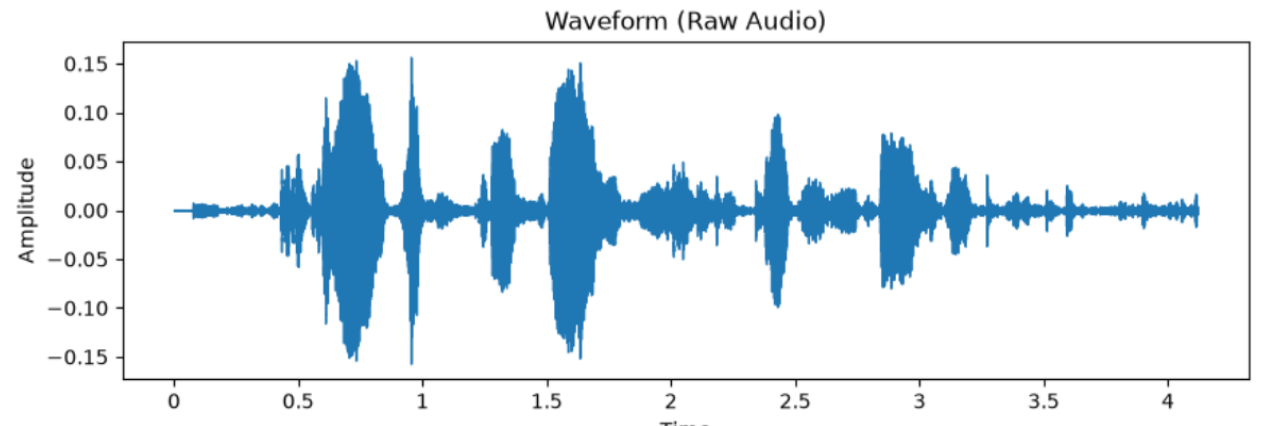
Introduction

What is SER?

- **Speech Emotion Recognition (SER)** analyzes emotions from audio signals
- Used in areas like customer service, mental health, and AI assistants
- Traditional systems focus mainly on text and often ignore vocal emotion
- **Goal:** build an end-to-end system in **Azure**
- Integrated with an **Azure AI Foundry** agent for interactive use
- Focused on both **dataset performance** and **real-world speech**

Problem

- Emotions in speech are complex and difficult to detect. Due to cultures, situations, individuals etc.
- Models perform well on **controlled dataset audio**
- Real-world audio varies (different speakers, accents, recording quality)
- In interactive AI systems (e.g. agents) this can lead to unreliable responses
- **Goal:** understand how well the model generalizes to real-world speech



Captures pitch, tone, energy, frequency patterns

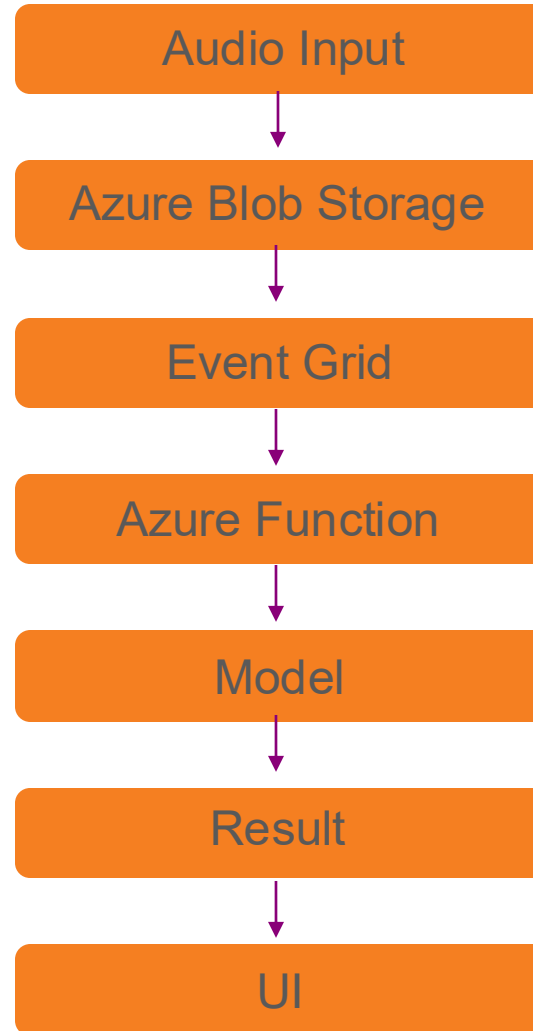
System Architecture

The Process

Backend: Azure Functions

Frontend: UI

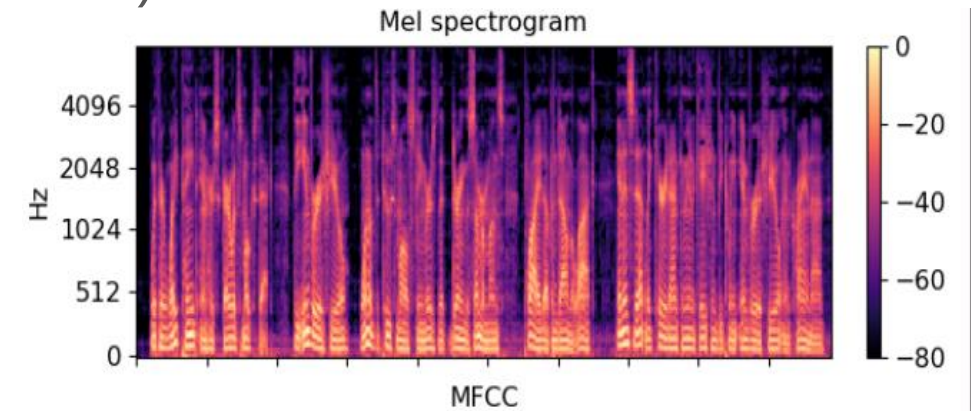
Model: SVM + MFCC



The predicted emotion can be used in an Azure AI Foundry agent to adapt responses.

Model Development

- Input: Audio → feature extraction
- Features: MFCC (Mel-Frequency Cepstral Coefficients)
- Model: Support Vector Machine (SVM)
- Output: Emotion (happy, sad, neutral, angry)
- Retrained with own recordings and additional speakers



The Pipeline:



AI Foundry Integration

- Azure AI Foundry enables building agent-based AI systems
- The agent can adapt its response based on detected emotion

Example:

Angry → Calmer response

Sad → Supportive response

In this project:

- Emotion is detected using a custom model
- Results can be applied in agent scenario
- Due to regional limitations, a custom UI was used for the demo

Demo

Emotion Agent

A conversational interface for retrieving speech emotion predictions from your Azure backend

How to use

1. Upload a WAV file to the backend system.
2. Ask the agent about that filename in natural language.
3. The agent retrieves the stored prediction and explains it.

Quick examples

demo_01.wav

record_h (1).wav

errored voice.wav



Hi! I'm your emotion agent. Ask me about the emotion of a processed WAV file.

For example: **What is the emotion of demo_01.wav?**



Live Speech Emotion Detection

Record your voice



..... 00:00

Results

Accuracy:
82.75%

Total samples:
2730

Correct
Predictions:
2259

	CREMA-D Dataset Table						
Actual / Predicted							
	Angry	Fearful	Happy	Neutral	Sad	Disgust	Total:
Angry	441	5	5	0	1	3	455
Fearful	16	423	2	0	13	1	455
Happy	13	26	407	4	2	3	455
Neutral	7	100	2	331	11	4	455
Sad	3	58	0	2	391	1	455
Disgust	7	92	10	27	53	266	455

Results

Accuracy:
70.45%

Total samples:
44

Correct
Predictions:
31

	Self-Recorded Data Table				
Actual / Predicted					
	Happy	Sad	Angry	Neutral	
Happy	8	0	3	0	
Sad	0	7	4	0	
Angry	1	0	10	0	
Neutral	0	1	3	6	

Accuracy drops in real-world recordings

Improvements

- Added real-world speech recordings
- Included multiple speakers for more variation
- Retrained the model with new data
- Improved performance on real-world input
- Reduced bias toward “angry” predictions

However, there are still limitations to be addressed...

Limitations

- Limited amount of real-world training data
- Real speech varies in speakers, accents, and recording quality
- Some emotions are still confused (e.g., predicted as “angry”)
- Model is a proof-of-concept not production-ready

Conclusion

- Built and end-to-end Speech Emotion Recognition system in Azure
- Integrated emotion detection into an AI agent using Azure AI Foundry
- Dataset performance: 82.75% accuracy
- Real-world performance: 70.45% accuracy
- Real-world speech remains more challenging than dataset audio
- Highlights importance of diverse and realistic training

Thank you for listening!



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